

# Milling Theory

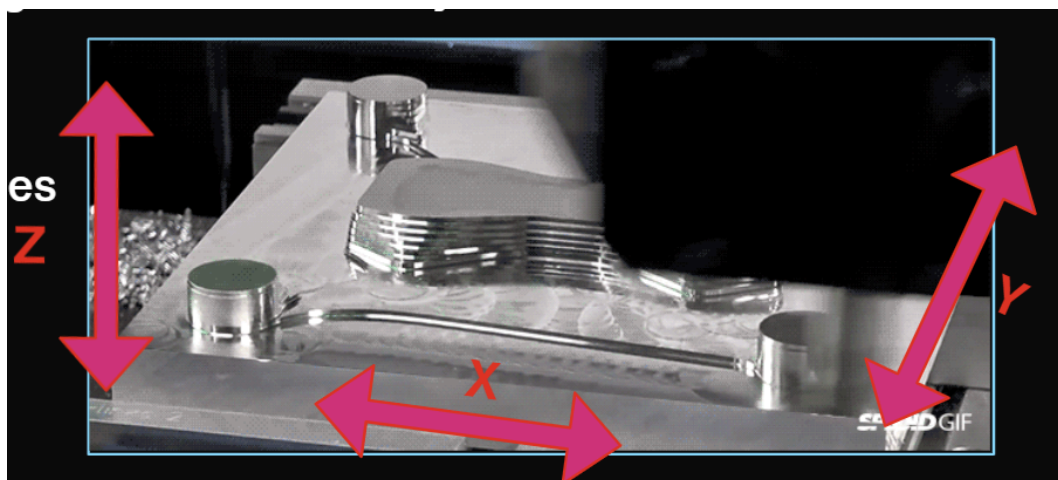
## Intro to the Milling Machine

A milling machine does a milling process to remove material.

A milling process is when work is fed into a revolving cutting tool to remove material. The cutting tool is usually a multi tooth cutter, which can be held horizontally or vertically.

The workpiece is then moved along the x y and z axis to feed the work into the tool.

The tool's height can also be adjusted as well.



This operation removes material at a faster rate than turning, and also can make a wider range of flat surfaces and parts unlike only circular parts.

There are different types of milling machines:

- Column and Knee Types

  - Vertical Milling Machine

  - Horizontal Milling Machine

  - Universal Horizontal Milling Machine

- Fixed Bed Type

- Planar Type

- Special Purpose Typed

These are the parts of a vertical milling machine:

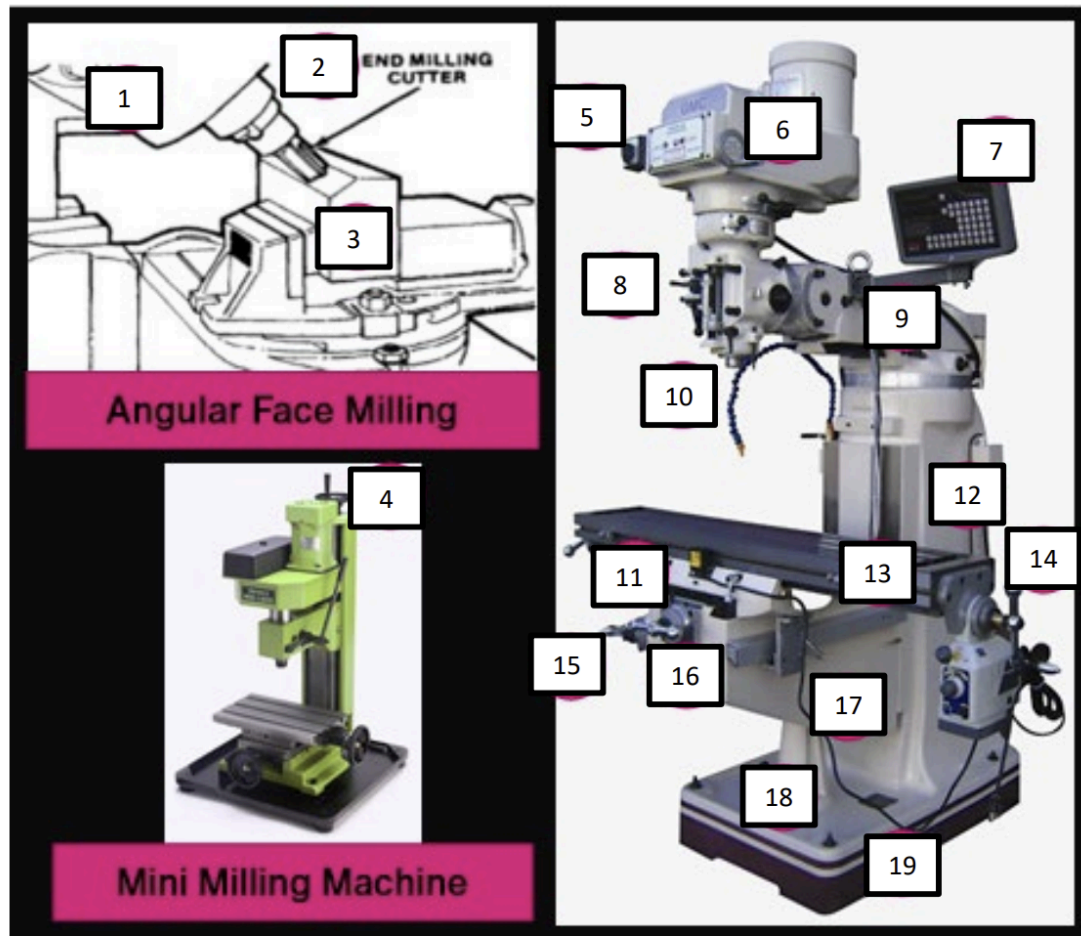


Figure 10-5 Vertical Milling Machine

|   |                       |                                                                                                                                                                                                                                          |
|---|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Motor Switch          | controls speed and direction of spindle                                                                                                                                                                                                  |
| 6 | Spindle Head          | contains the motor, gearbox, drawbar, draw clamp, spindle, quill and bearings<br>motor powers spindle,<br>gearbox manages spindle speed & torque ratio,<br>quill allows spindle to be lowered<br>whole assembly can be tilted vertically |
| 7 | Digital Readout (DRO) | Indicates location of work along X, Y and Z axis, allowing it to reset position and for accurate cutting.                                                                                                                                |
| 8 | Quill Feed            | quill lowers spindle, useful for drilling<br>quill can be fed vertically via a handle or handwheel for finer movements.                                                                                                                  |

|    |                              |                                                                                                                                                                                                                                                                     |
|----|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Ram/Overarm                  | holds motor and spindle head, allowing it to be extended, retracted and swivelled horizontally                                                                                                                                                                      |
| 10 | Collet Chuck                 | where cutting tools are mounted (onto the collect chuck), often with a spring collet (type of collet chuck???)<br>The chuck is connected directly to the spindle, via a drawbar clamp or drawbolt inside the spindle, hence allowing it to rotate along the Z axis. |
| 11 | Saddle                       | mounted on horizontal guideways on the knee, can also towards or away from the column.<br>Houses lead screw nut to move table horizontally (???? counter intuitive)                                                                                                 |
| 12 | Column                       | Rear of the main body; houses electrical circuitry while supporting the ram.<br>has guideways in front for knee to be moved vertically                                                                                                                              |
| 13 | Table                        | mounted on saddle guideways (perpendicular to knee guideways)<br>table can move left and right via lead screw                                                                                                                                                       |
| 14 | Feed Handle                  | turns lead screw to move table left and right<br>may have motor attachment to drive table automatically                                                                                                                                                             |
| 15 | Vertical Traverse Hand Crank | turns screw jack when rotated, causing knee to move vertically                                                                                                                                                                                                      |
| 16 | Cross Traverse Handle        | turns lead screw on knee when rotated, moving saddle in forward/backward direction                                                                                                                                                                                  |
| 17 | Knee                         | Mounted on vertical guideways on front side of column; for it to be moved vertically along column<br>encloses a screw jack for knee to move vertically accurately, as well as enabling table to move vertically<br>carries saddle table and other acessories        |
| 18 | Screw Jack                   | Allows knee to move vertically, when rotated(via vertical traverse hand crank)                                                                                                                                                                                      |

|    |                              |                                                                                                                               |
|----|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 19 | Base                         | supports machine<br>some bases have filters to filter out chips and recycle coolant.                                          |
| 1  | Swivelled Spindle Head       | The spindle head can be swivelled for angular faced milling                                                                   |
| 2  | End Mill [Cutting Tool]      | A type of cutting tool attached via the collet chuck?? for angle faced milling????                                            |
| 3  | Workpiece Clamped with Vice  | A vice on the table clamps down the workpiece                                                                                 |
| 4  | Vertical Traverse Hand Wheel | unlike the vertical traverse hand crank, it moves the motor and spindle vertically, by rotating the lead screw in the column. |

These are the parts of a horizontal milling machine

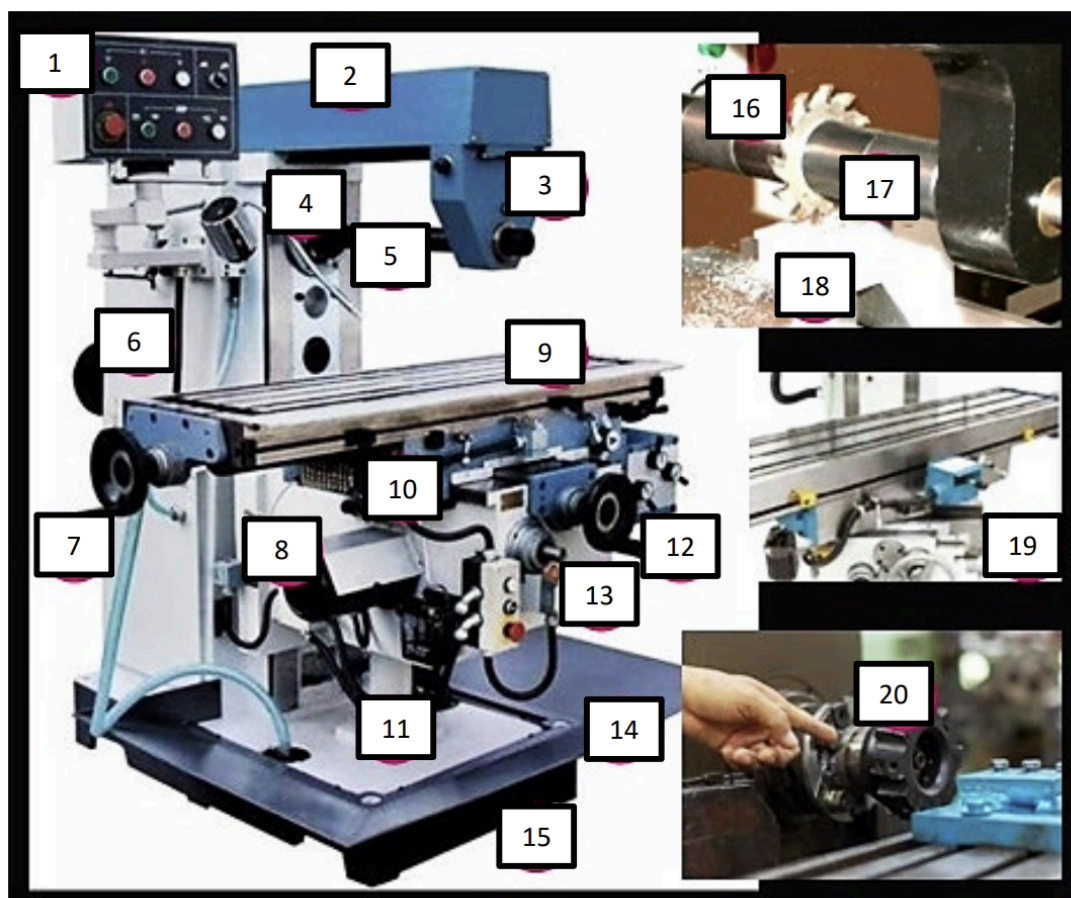


Figure 10-3 Horizontal Milling Machine

|   |               |                                            |
|---|---------------|--------------------------------------------|
| 1 | Control Panel | controls spindle speed and other functions |
|---|---------------|--------------------------------------------|

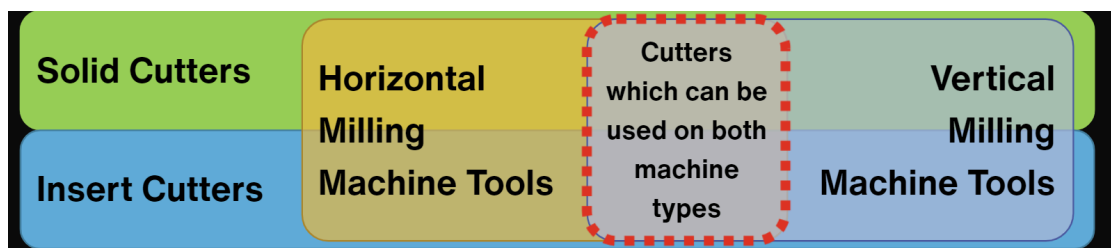


|    |                         |                                                                                                                                                                 |
|----|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2  | Overarm                 | Supports free end of arbor<br>Can be extended and retracted, depending on size of arbor                                                                         |
| 3  | Arbor Support           | also supports free end of arbor                                                                                                                                 |
| 4  | Spindle                 | Connected to motor, allows arbor/stub to be connected to it.                                                                                                    |
| 5  | Arbor                   | Allows cutting tools & spacers to be mounted on it<br>connected to spindle directly via a drawbolt/drawbar clamp in spindle<br>rotates horizontally             |
| 6  | Column                  | At rear of main body; holds circuitry, motor, spindle, and spindle bearings                                                                                     |
| 7  | Feed Handle             | turns lead screw when rotated, moves table in left/right direction (longitudinally)                                                                             |
| 8  | Vertical Traverse Motor | turns screw jack when rotated, moving knee vertically. Hand crank may used instead of motor                                                                     |
| 9  | Table                   | mounted on saddle<br>guideways(perpendicular to knee guideways)<br>By lead screw rotated by feed handle, it moves in left/right direction                       |
| 10 | Saddle                  | Mounted on horizontal guideways on knee, allowing front/back movement (via another different lead screw)<br>houses lead screw nut to move table                 |
| 11 | Screw Jack              | Allows knee to move vertically when rotated                                                                                                                     |
| 12 | Cross Traverse Handle   | turns lead screw on knee to move saddle in front/back direction                                                                                                 |
| 13 | Knee                    | mounted on vertical guideways on column; moves vertically<br>encloses a screw jack for raising itself accurately<br>carries saddle, table and other accessories |
| 14 | Chip Pan                | Collects falling chips and coolant                                                                                                                              |
| 15 | Base                    | supports machine<br>some bases have filters to filter out chips                                                                                                 |

|    |                                             |                                                                                                                             |
|----|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|    |                                             | and recycle coolant.                                                                                                        |
|    |                                             |                                                                                                                             |
| 16 | Side & Face Staggered Cutter [Cutting Tool] | Cutting tools, like the Side & Face Staggered Cutter is mounted on the arbor                                                |
| 17 | Spacers                                     | Positions and locks cutting tool along arbor                                                                                |
| 18 | Workpiece Clamped with Vice                 | A vice on the table clamps down the workpiece                                                                               |
| 19 | Swivel Table                                | **for UNIVERSAL HORIZONTAL MILLING MACHINE,<br>it has a swivel table, which can be rotated up to 45 degrees for angled cuts |
| 20 | Face Mill [Cutting Tool] on a Stub Arbor    | Cutting tools, like the Face Mill, can be attached to a Stub/Short Arbor, where there is no support on the free end.        |

## Milling Cutters

There are different kinds of cutting tools used, which can be categorised depending on which type of milling machine(s) its suited for, and whether its a solid or insert cutter.



this "venn diagram" shows the categories of tools, and also shows that some tools can be used for both horizontal and vertical milling

## Solid Cutters

Solid Cutters have their cutting edges part of the cutter body.

Thus, the cutting edges are longer, as they make up more of the tool length, hence allowing deeper cuts.

usually High Speed Steel (HSS) is used, and higher quality cutters may be coated with alloys to resist wear, reduce friction and heat.

Highest quality cutters are made with carbide, with coatings, allowing them to withstand higher cutting forces and heat, despite the high price

### Insert Cutters

These cutting tools are 2 separate parts, the insert and the holder, with the inserts locked onto the holder via the wedge or screw method

The cutting length depends on the number of inserts

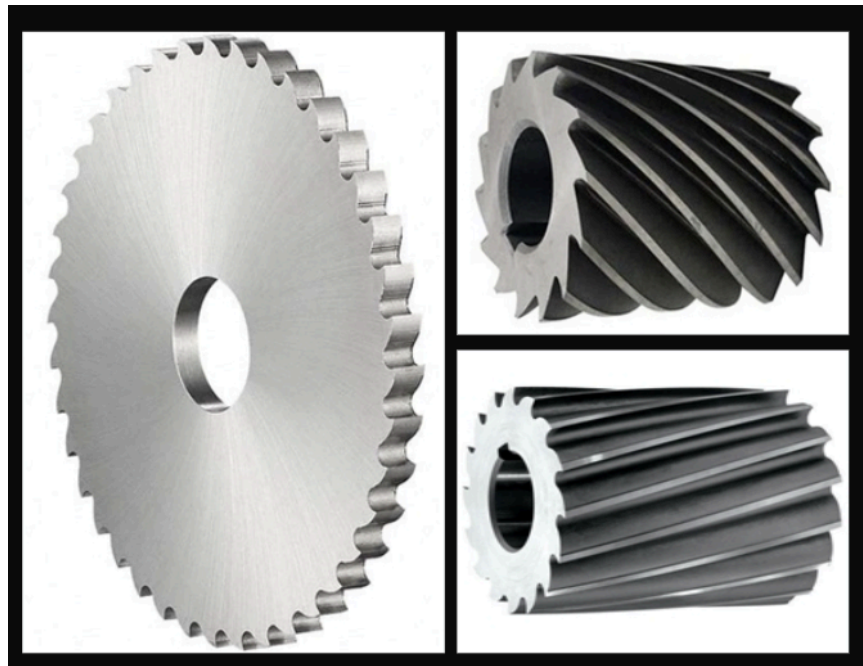
Recall that inserts are indexable(multiple cutting edges in 1 insert) and are economical (due to easy to be discarded) from lathe theory (hence same applies for milling)

Usually, tungsten carbide is used for inserts.

### Horizontal Milling Machine Tools

There are different types of horizontal milling tools, with different shapes, sizes and number of teeth.

The cutting teeth can also be straight, or helical, with helical type cutters being divided into right hand tools (RHT) or left hand tools (LHT), based on whether the spiral turns clockwise, or counter clockwise.



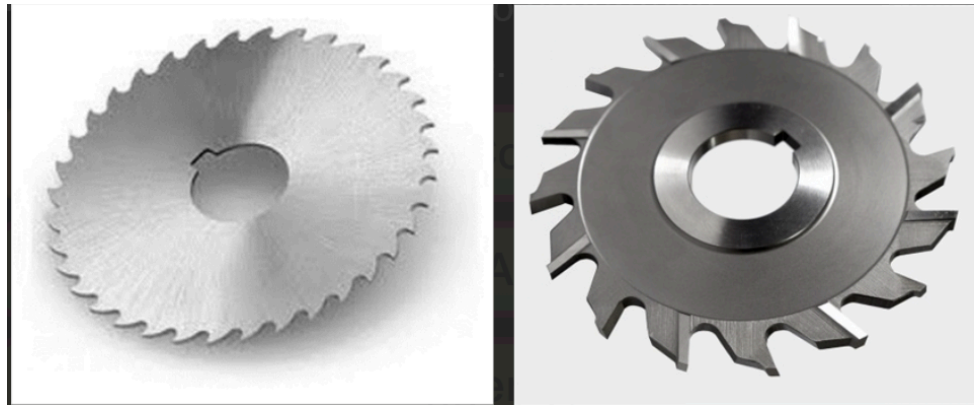
All horizontal milling tools have a keyhole to allow it to rotate with the arbor

Most horizontal milling tools are form relieved tools, meaning they create the shape they are designed to cut

listed are some types of horizontal milling tools

#### Slitting Saw

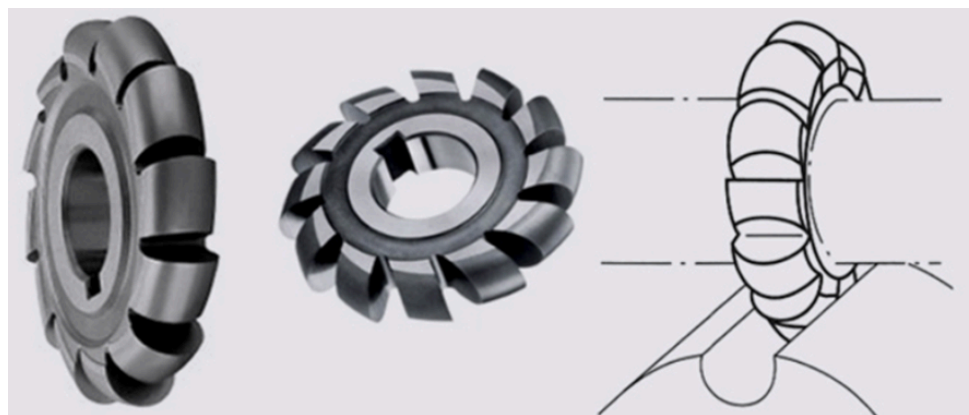
comes in straight or staggered flutes(flutes meaning grooves); aka the cutting edges are either straight (parallel to each other) or are tilted, with the edges alternating between left and right.



These are useful for milling narrow slots, parting, or cutting recesses at corners

#### Convex/Concave Cutter

A convex cutter has convex "blades", useful for creating internal radii



while a concave cutter has concave "blades", useful for creating external radii.

## Concave Cutter



## Plain/Slab Mill

A plain/slab mill is used for machining flat surfaces on the outside diameter, held with a long arbor

These mills are usually helical, available in LH or RH spiral, with different helical angles used depending on the material cut



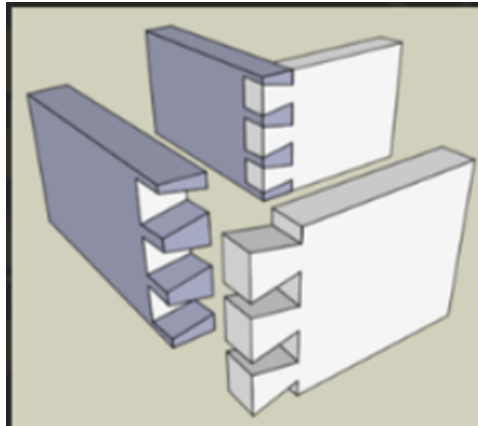
Right Hand Mill

Left Hand Mill

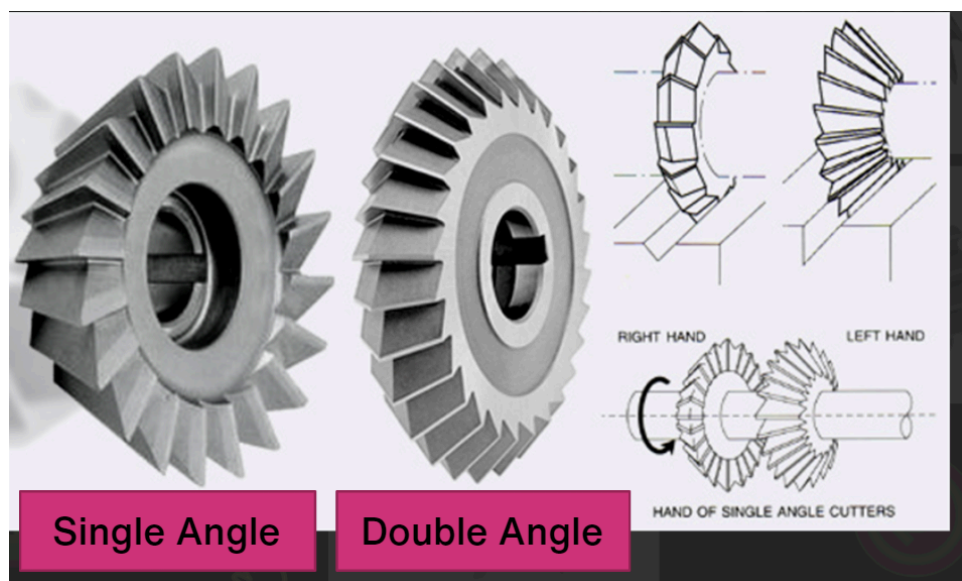
## Angle Cutter

Has tapered off cutting teeth, similar look to a bevel gear  
useful for chamfering, or for dovetail processes.





These cutters come in either single angle, or double angle cutters (similar to a single and double sided bevel gear)



Helical angle cutters are also used, with LH or RH spirals

#### Side & Face Staggered Cutter

These cutters have teeth staggered left and right, causing each tooth to cut a smaller area, which reduces cutting forces

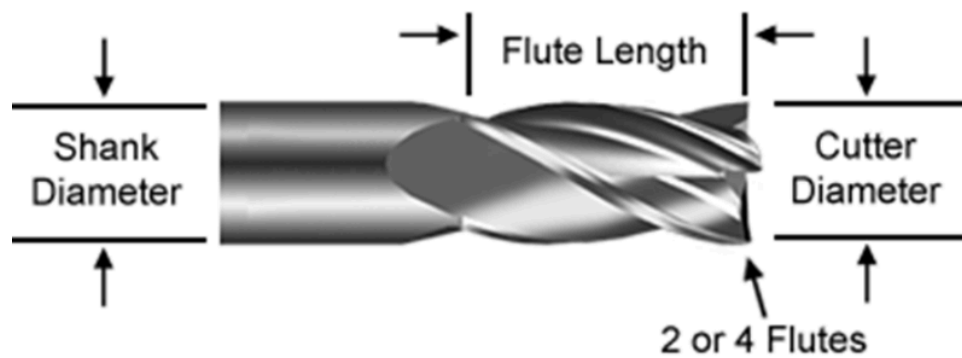
Thus, these cutters are used for machining vertical and horizontal surfaces simultaneously, making it useful for slots or stepped surfaces



### Vertical Milling Machine Tools

Again, there are different types of vertical milling tools, with different shapes, sizes and number of teeth, with the teeth being straight or helical.

Vertical Milling tools can be identified if they have a shank (blank shaft portion), as they need to be mounted onto the spindle

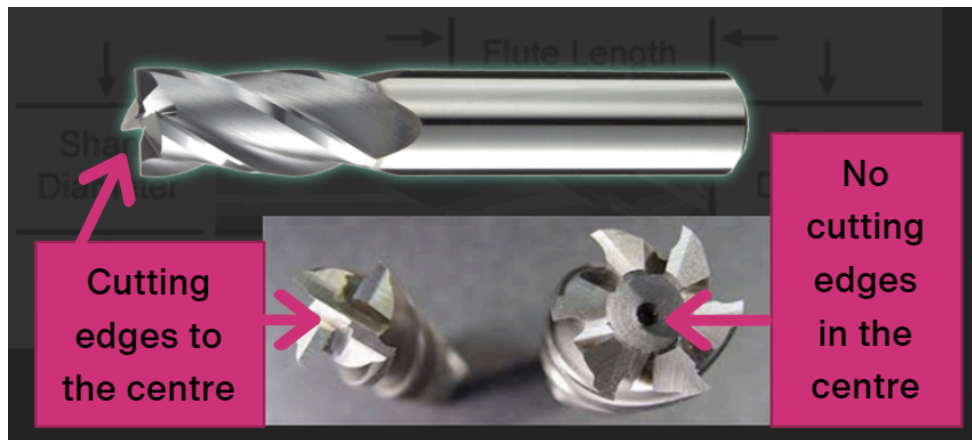


Some vertical milling machine tools are:

#### End Mill

This tool is mounted with the spring collet and collet chuck.

Used for milling flat/stepped surfaces, or open ended slots or keyways. Some end mills can also be used for plunge milling, only if it has cutting edges to the centre of the tool



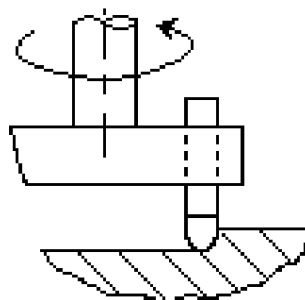
These come in LH and RH spirals, with different spiral angles for different work materials.

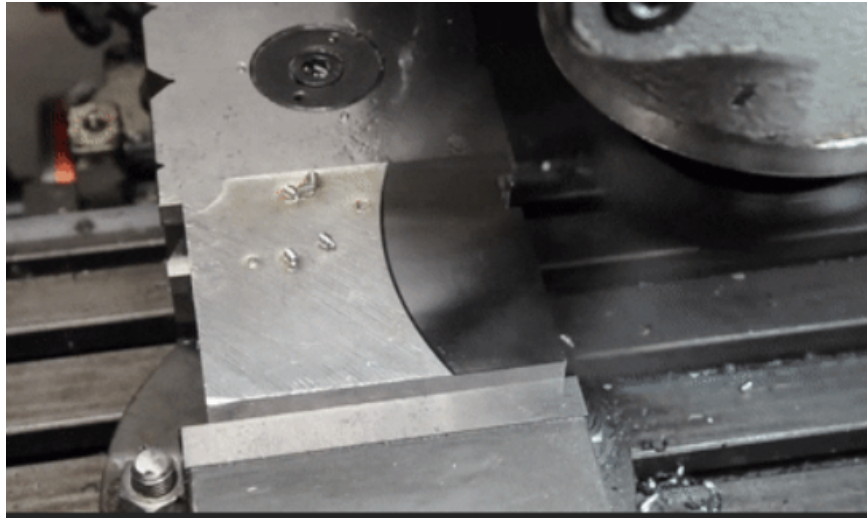
\*\*open ended slots is when the slot goes through the whole length of the workpiece.

#### Fly Cutter

This tool is mounted with a holder, to the spring collet and collet chuck.

These cutters have a single point tool for milling large flat surfaces.





the tool length (in this case how far the single point tool is from the rotation axis) can be adjusted, to adjust the cutter's effective diameter

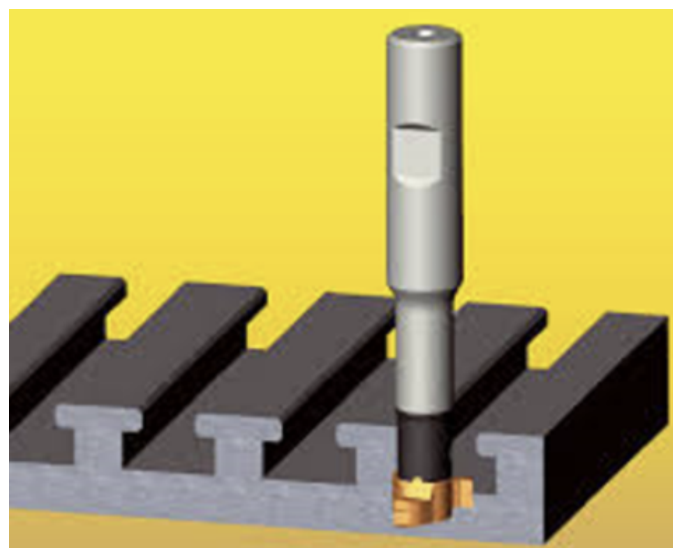
This method of cutting flat surfaces compared to face mills, is cheaper, but less productive

\*\*there are somewhat similar to adapters for turning tools from the lathe.

#### T Slot Cutter

This tool is mounted with the spring collet and collet chuck.

this tools has cutting edges on its outside diameter, for cutting horizontal T shape slots, useful for machine tables (as sliding rails), for easy installation



Before using this tool, a vertical slot, making up the vertical portion of the T, has to be milled first, large enough for the neck and shank of the T slot cutter to have enough clearance and fit through to cut the workpiece.



### Face Mill

This tool is mounted via an arbor on the spindle

The tool's cutting edges are on its outside diameter and bottom end are used for milling large flat surfaces

These tools are available in different diameters.



### Ball Nosed End Mill

This tool is mounted with the spring collet and collet chuck.

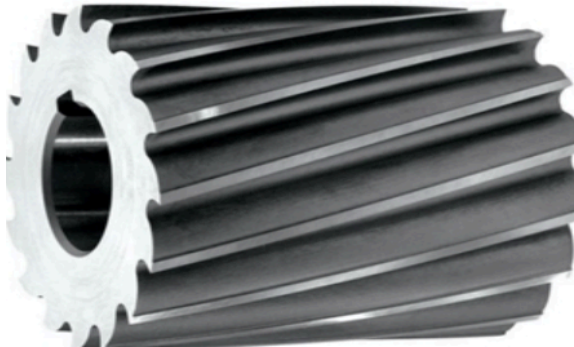
The tool has a ball shape at its end, and with the cutting edges on its outside diameter and bottom end, its used for milling cavities and concave surfaces.





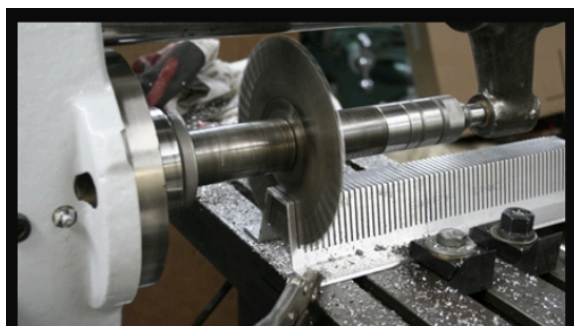


**Plain Milling**



**Saw Milling**

Creates slits in the workpiece  
done with a slitting saw



**Saw Milling**

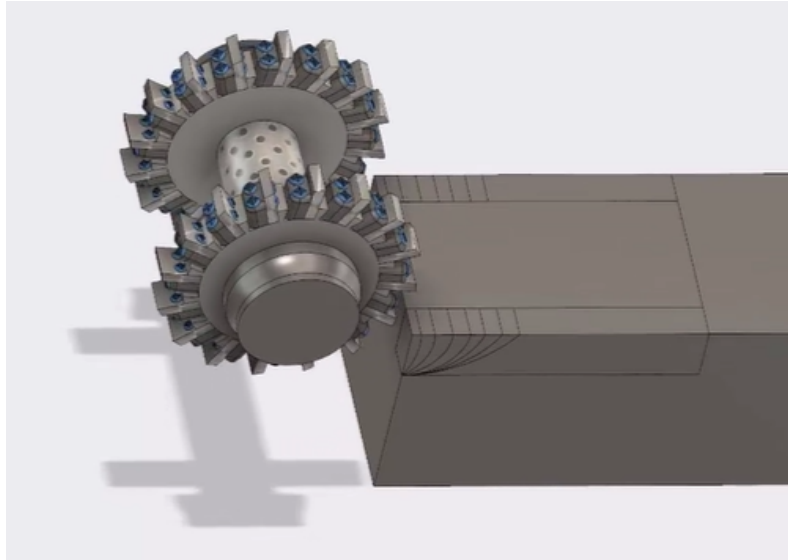


There are also more special/rare milling operations:

#### Straddle Milling

Creates parallel surfaces of same depth

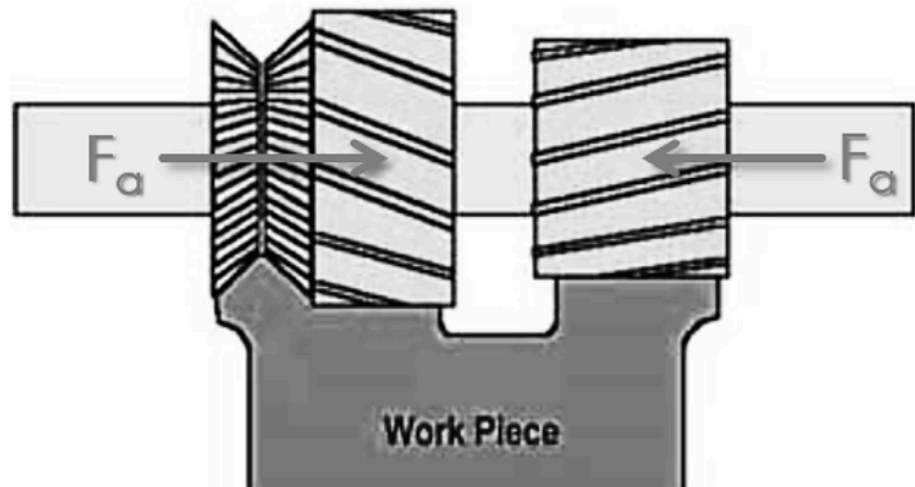
done with 2 equal sized face and side cutters, mounted on the arbor with spacers to position the cutters away from each other.



#### Gang Milling

Creates a combination of formed surfaces (e.g. angular, flat, curved) without changing tools, either on one or multiple workpieces

Different cutters are mounted together on the arbor depending on the desired surfaces, however a combination of LH and RH cutters must be used to cancel out axial forces, preventing stress and damage.



## Vertical Milling

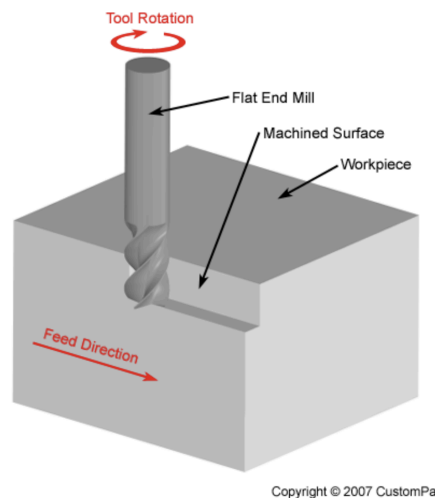
### Operations w/ the End Mill

Recall the end mill tool from earlier. It can do many operations, such as

#### Contouring/Side Milling

using the cutting edges on the side, it can size the workpiece or create a desired shape on the work.

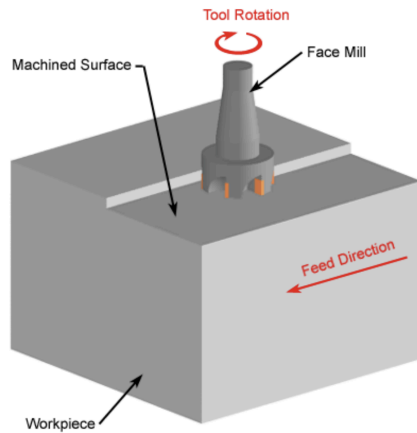
This is analogous to the turning operation on the lathe.



#### Facing

using the cutting edges on the bottom, it can remove material from the entire top surface of the work.

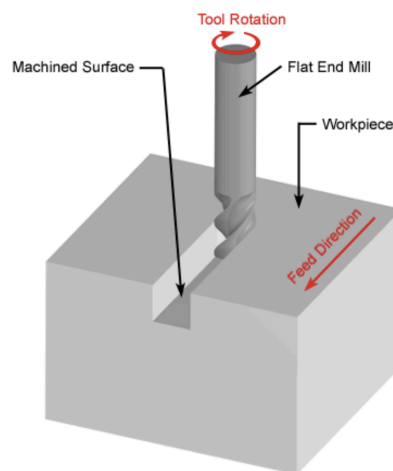
This is analogous to facing on the lathe



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## Slot Milling

using cutting edges on both the sides and bottom, it creates slots in the workpiece

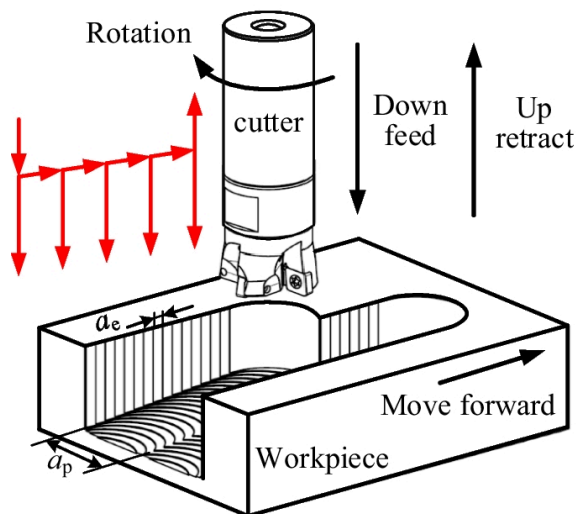


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## Plunge Milling

removes material from the workpiece in a fashion similar to drilling (fed downwards), however without using a drill bit.

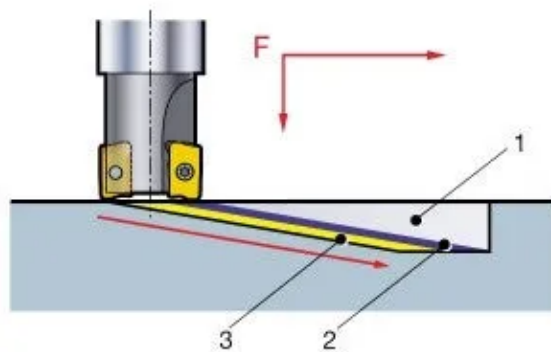
as said earlier, this operation requires cutting edges on the bottom which extend to the center of the tool



### Ramping

An alternative plunge milling; i.e. it still removes material similarly to drilling, however is suited when the end mill does not have cutting edges at the bottom that extend to center.

Instead, the tool is fed downwards and sideways, to use the bottom section of the outer cutting edges to remove material, hence forming a ramp as it does so:



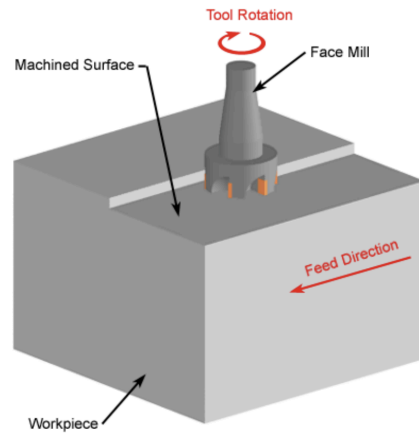
### Other Common Operations

#### Facing

remove material from the entire top surface of the work

face mill is used

a face mill of smaller diameter than the workpiece requires multiple cutting passes to fully face the work.

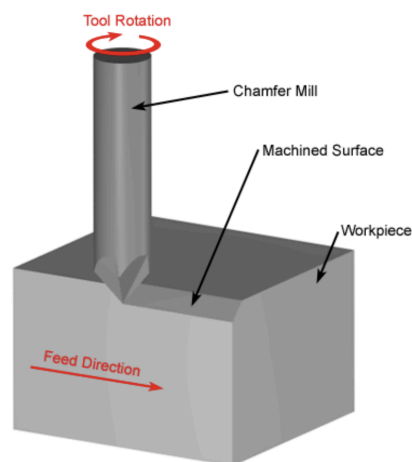


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## Chamfering

removes material from the edges, creating angled surfaces

chamfering tool is used



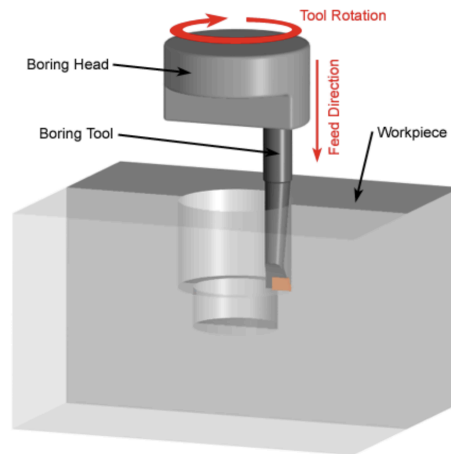
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## Boring

removes material from existing hole to enlarge it

boring tool is used

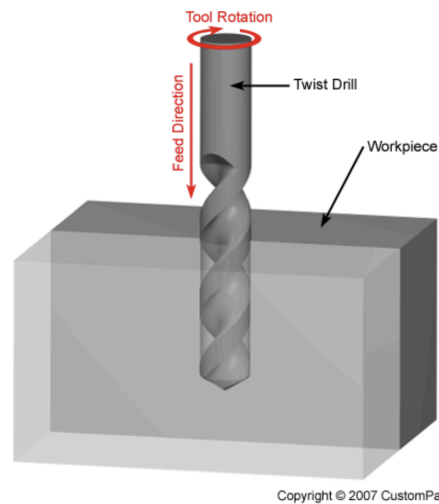




## Drilling

creates a hole in the workpiece

drill bit is used

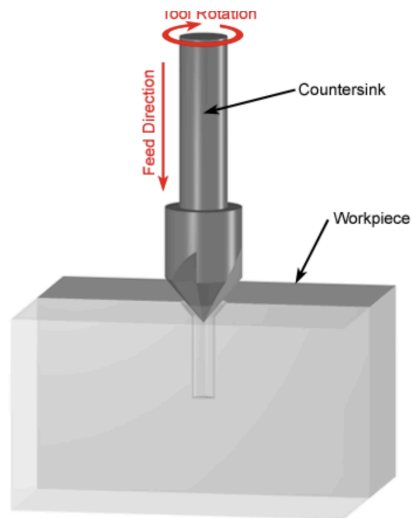


## Countersinking

creates a countersunk hole from an existing hole,

countersinking tool is used

useful for making countersunk head screws flush with the top surface

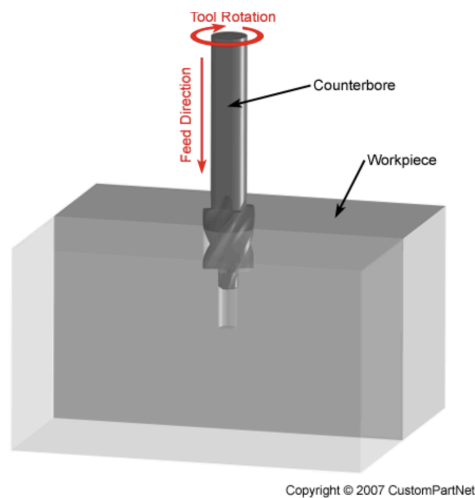


## Counterboring

creates a counterbore hole from an existing hole

counterboring tool is used

useful for making counterbore head screws flush with the top surface

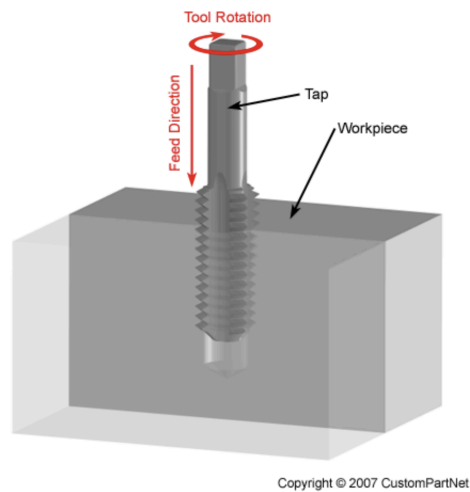


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## Tapping

creates internal screw threads from an existing hole

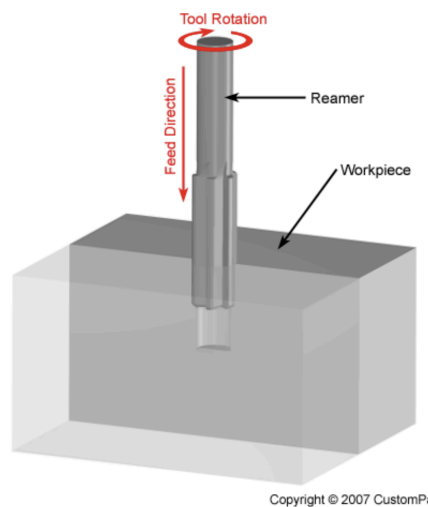
thread tap is used, with high amounts of lubrication



## Reaming

finishes and creates accurate holes

reaming tool is used, at low speeds with high amounts of lubrication



## Milling Fundamentals

### Cutting Methods

Dependent on the cutter orientation, and the direction of cut.

### Peripheral Milling

Cutter axis is parallel to work surface, hence only the side/circumference of the cutter is used to machine the work.

This causes a wave like surface finish (rows of lines along axis of tool rotation)

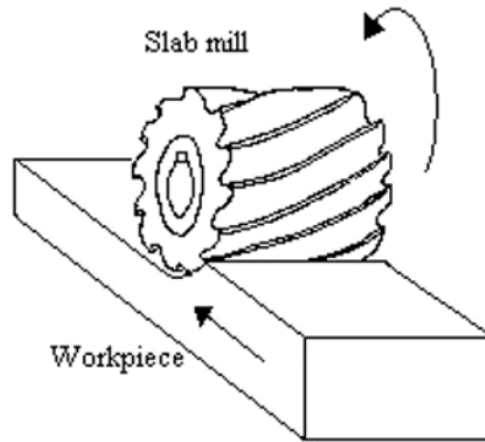
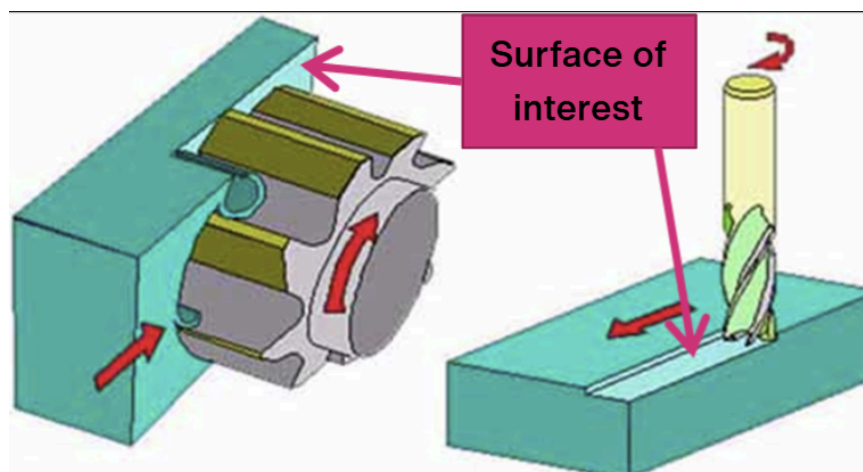


Figure 11-2 Wave Like Surface Finish

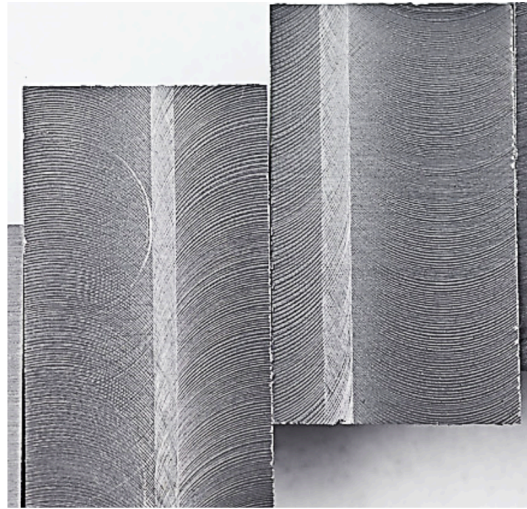
### Face Milling

Cutter axis is perpendicular to work surface, however circumference still does main cutting, while the end of the cutter finishes the work, by removing a small amount of material, affecting the surface finish by causing circular marks on the work.



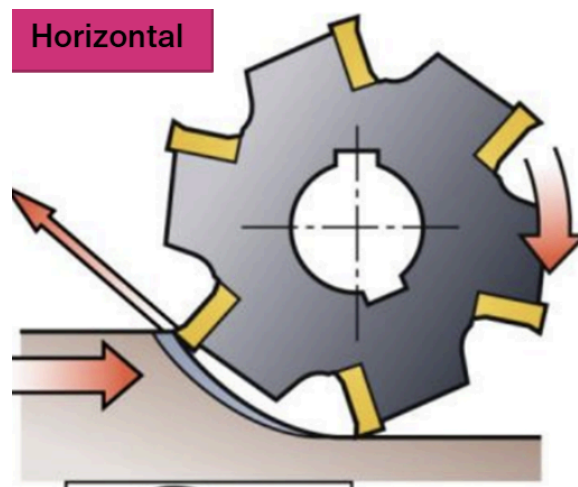
Notice how (on the left) the side cutting edges cut and remove the chips.

The bottom of the tool will do surface finishing, producing circular marks:



### Up Cut/Conventional Milling

Up Cut refers to when the work is fed against the cutter:



#### Advantages:

- negates backlash error, as tool does not push work against clearance, making it useful for older milling machines

#### Disadvantages:

- high feed power needed as work is pushed away
- chip welding on cutting edges at end of cut, due to chips growing from 0 to its maximum (as shown above) and high temperatures.

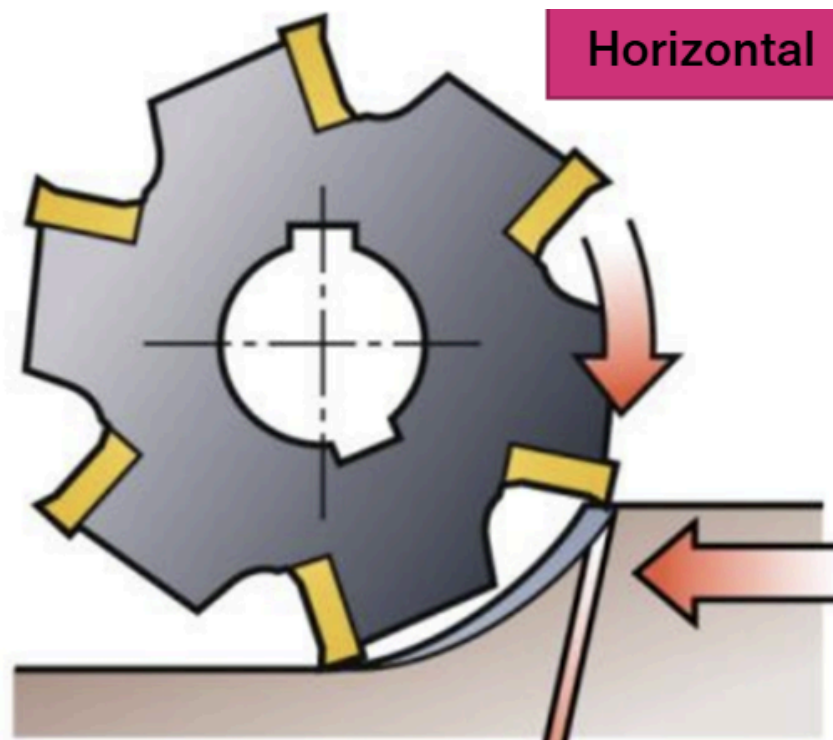
Poor finish, as chips are carried by cutter onto the newly cut surface, scratching it

cutter wears faster, as it initially rubs against work

for horizontal milling, the work tends to be lifted up

#### Down Cut/Climb Milling

Down cut refers to when the work is fed in the same direction as the cutter



#### Advantages:

Lower feed power needed as work is fed with cutter

Cutter can be well lubricated/applied with coolant

Hence no chip welding, as chip thickness decreases from maximum to 0 ( ) + low temperature at end of chip from coolant

Good finish as well, as chips are deposited behind cutter and do not scratch the newly cut surface.

Work is pushed down by tool, improving clamping

#### Disadvantages:

Backlash effects are more pronounced, as it can push the workpiece into the clearances, causing dimensional inaccuracy.

Hence newer machines use down cutting instead.

#### Backlash (in up/down cut milling)

Most parts in the milling (and also lathe) machines use a nut on a leadscrew to move components. As such, there needs to be clearance between the two for the nut to move.

However, this causes imprecise movement, as the nut can still move even when the leadscrew doesn't. This is called backlash, and worsens with more wear and tear.

One way to prevent this is by using a recirculating ball screw, where ball bearings reduce the clearance between parts and hence reducing backlash.

#### Cutting Factors

Like turning, milling has factors affecting cutting as well, some which overlap with turning.

##### Workpiece Material

#### **[SAME AS LATHE THEORY]**

Depending on the part needed, different materials with different properties (composition of material, manipulated mechanical properties like heat treatment) may be used, thus requiring different tools, cutting conditions and machine power.

#### Cutting Conditions

Like lathe theory, there are 3 cutting conditions

Cutting Speed (dependent of how fast the TOOL spins, not the work)

Same as lathe theory, however with the tool spinning instead. So,

Cutting speed refers to tool surface speed, not work surface speed

working range is determined by type of tool, workpiece material (same as lathe) AND tool diameter.

Again, spindle speed is not the same as tool surface speed.



For rougher/less precise cuts, a lower cutting speed is used  
For finer/more precise/surface finishing cuts, a higher cutting speed is used.

Feedrate (of the WORK into the tool, not the tool into the work)

### **[SAME AS LATHE THEORY]**

#### Depth of Cut

Same as lathe theory (i.e. working range depends on type of tool and work material) however cutting length used on tool also affects depth of cut working range.

Again, rougher cuts → higher depth of cut, finishing cuts → lower depth of cut

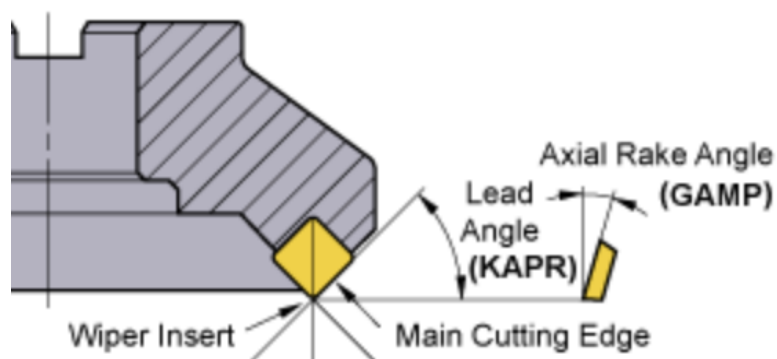
#### Tool Material & Characteristics

### **[SAME AS LATHE THEORY]**

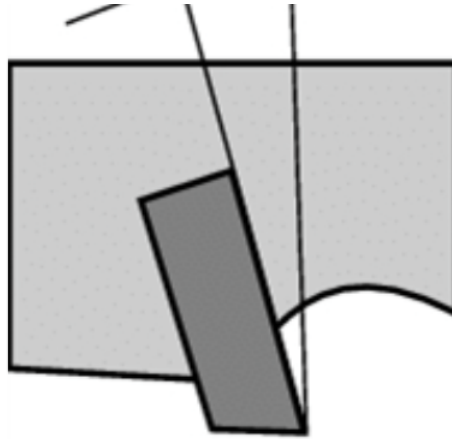
#### Tool Geometry

Instead of 6 angles for single point cutting tools, there are only 2 main angles for milling tool geometry, Radial Rake and Axial Rake Angle.

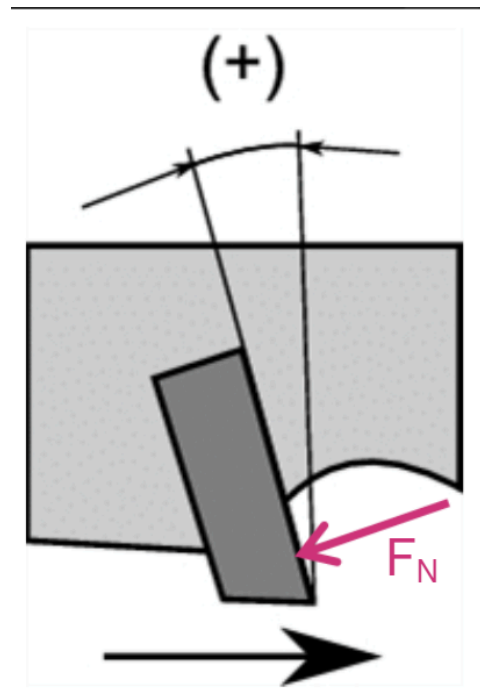
Radial Rake is measured from the tool to the radius, like so,



while axial rake angle is measure from the tool to the axis,



The Axial Rake Angle will be more focused on,  
Positive (Axial) Rake Angle



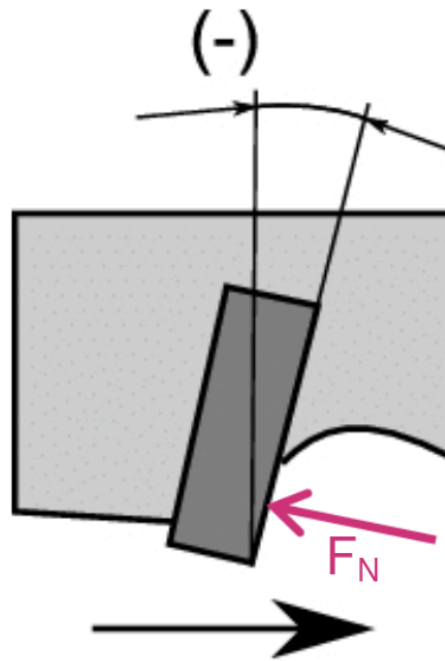
Recall from lathe theory that a positive rake angle is when the surface slopes back.

Hence,

Normal force of work on tool acts on an unsupported tool tip, making tool more likely to break.

Lower cutting force is needed as material flow is more fluid (and also better surface finish???)

Negative (Axial) Rake Angle



A negative rake angle is achieved when the surface slopes forward.

Hence,

normal force acts on supported tool tip, making tool less likely to break/stronger, permitting higher cutting speeds.

negative rake angle also generates more heat, causing more plastic chips.

#### Cutting Fluids

**[SAME AS LATHE THEORY]**

#### Machine Design & Condition

**[SAME AS LATHE THEORY]**

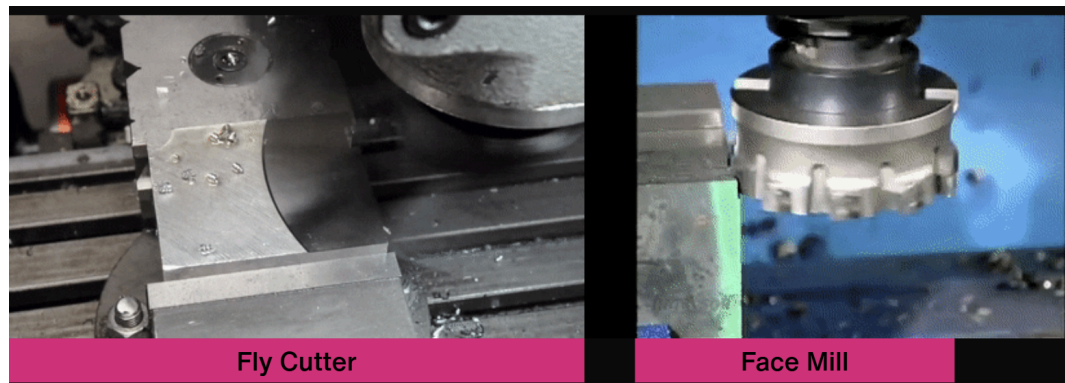
Depending on the type of lathe (e.g tabletop mill, industrial mill) have different powers and different purposes.

appropriate use and repair frequency affects the machine condition.

#### Number of Cutting Teeth

The number of cutting teeth is “directly proportional” to the feedrate, as each tooth cuts a set amount of material, so more teeth can cut more material, and hence can be fed faster to cut more material.

So a fly cutter, with 1 cutting tooth, needs to be fed slower than a face mill, with multiple teeth.

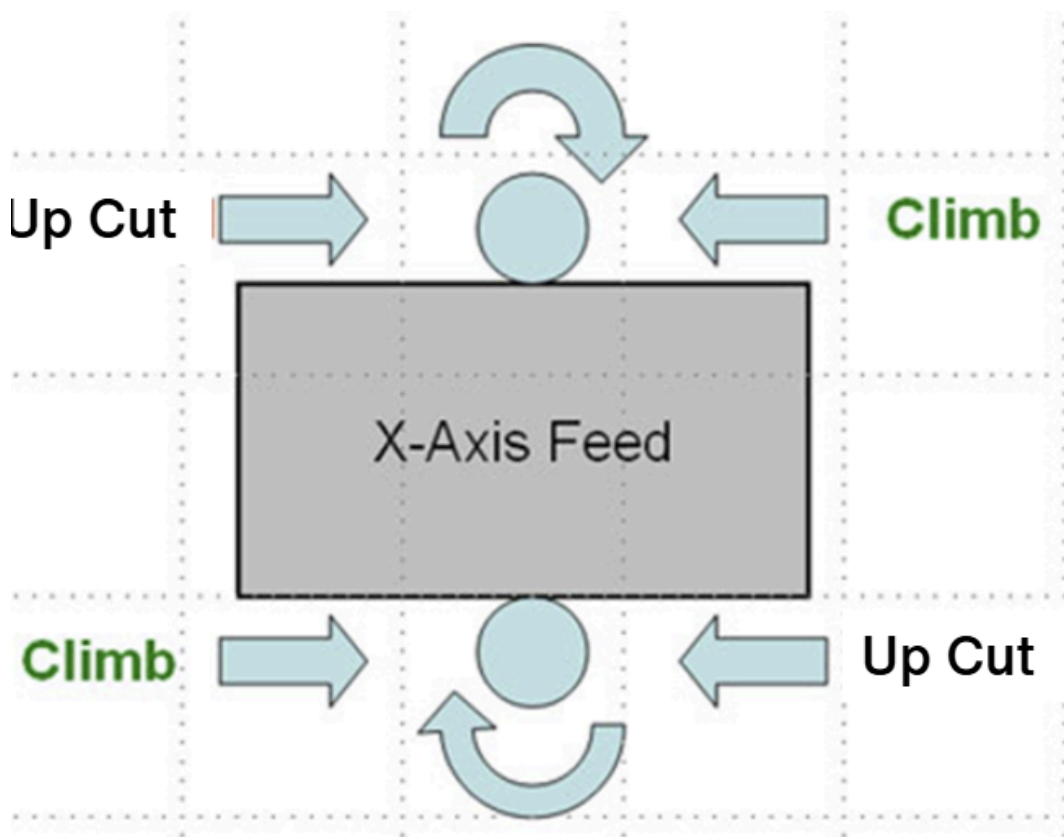


### Direction of Cut

depends on the tool rotation direction and the feed direction of the workpiece at a "point" where the tool and feed contact.

If both flow together, it is undergoing down cut, while if they flow against each other, it undergoes up cut.

This affects the finish and other factors said earlier.



One thing to note is that a workpiece may undergo both up and down cutting at the same time, like when the workpiece undergoes plunge milling, and is fed out to form a close ended slot.

The workpiece will then undergo both up and down cutting.

## Feeds, Speeds, and Machining Time

### Cutting Equations

Recall cutting speed ( $V$ , m/min) and spindle speed ( $N$ , rpm). The same equation is used to relate the spindle and cutting speed. In this case,  $D$  is the diameter of the tool, not the work.

$$V = \frac{\pi D N}{1000},$$

$$N = \frac{1000 V}{\pi D}$$

Feedrate can also be calculated, however there are now 3 forms for feedrate,

how much the work advances per rev, in mm/rev  $[F_r]$ , how much the work advances per min, in mm/min  $[F_m]$  and how much the work advances per tooth, in mm/tooth  $[F_t]$ .

The conversion formulas are as so: (first one is same, second is with mm/tooth, where  $z$  is number of teeth)

$$F_m = N \times F_r$$

$$\left[ \frac{mm}{min} = \frac{rev}{min} \times \frac{mm}{rev} \right]$$

$$F_r = z \times F_t$$

$$\left[ \frac{mm}{rev} = \frac{teeth}{rev} \times \frac{mm}{teeth} \right]$$

( $z$  can be thought of as teeth per revolution, as we count the number of teeth across the whole circumference of the tool, so  $z$  is in teeth/rev, where rev = 1)

Using the feedrate, the cut time can be calculated.

If the tool needs to advance  $l$  mm to make a full cut (e.g.  $l$  = length of workpiece for turning, or  $l$  = radius of workpiece for facing), the equation for cut time is

$$\text{Cut time} = \frac{l}{F_m}$$

where the cut time is in minutes.

## Tool & Work Holding Devices

There are different ways to hold the milling tools, like arbors, stub arbors, drill chucks, and spring collet & collet chuck.

### Spring Collet & Collet Chuck

Used to hold straight shank tools, e.g. an end mill



spring collet is placed in chuck → collet nut tightens to collet chuck → compresses spring collet → allows shank to be placed inside collet and held.

The collet nut is tightened/loosened via a C spanner.



The way the spring collet is placed in the chuck is that its first fixed onto the nut by placing it at an angle, and is removed at an angle as well.

There are differently sized collets, for different tool shanks, so that the same collet chuck can be used, as the collet chuck is fixed onto the spindle.

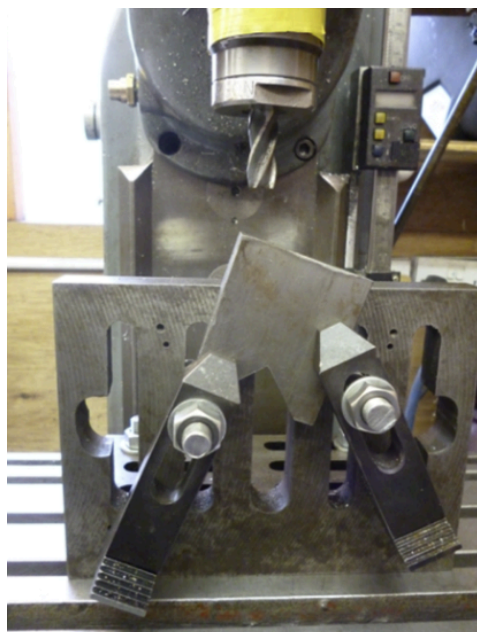
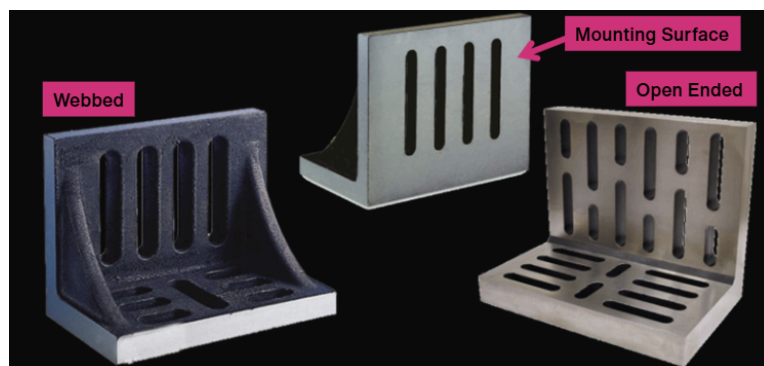
For milling, there are multiple work holding devices used, to clamp it down securely without damaging the work surface.

### Angle Plate

sets up workpiece for machining at 90 degrees to table.

the angle plate is bolted onto the table, while the work is mounted onto the mounted surface, callibrated to be perpendicular.

there are 2 types of angle plates, webbed and open ended.



the workpiece is then held on the mounting surface via clamps.

### Vee Block



holds shafts/cylindrical components, for milling keyways, slots, or flat surfaces.

they help to prevent bending under cutting pressure and prevent damages from clamping pressure.

different sized vee blocks may be used depending on the work size and use.

## Vices

quickest and most efficient method for holding parallel workpieces/regular flat surfaces

vices have a crank handle to quickly lock and release the workpiece, by moving the movable jaw.

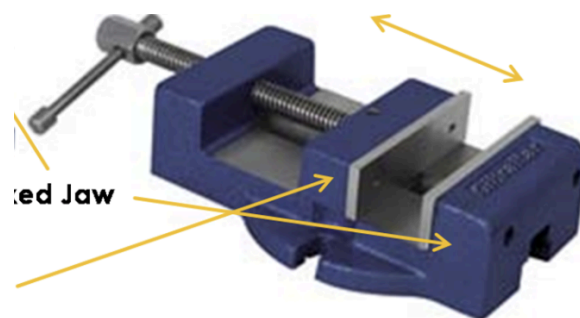
vices also have a fixed jaw, which can be used as the locating datum for workpieces to be aligned to, useful for the DRO to give the correct readings.

when using a vice, cutting forces should be directed to the fixed jaw, to reduce vibration and prevent the forces from loosening it.

hydraulic vices may also be used for higher clamping forces.

### Plain Vice

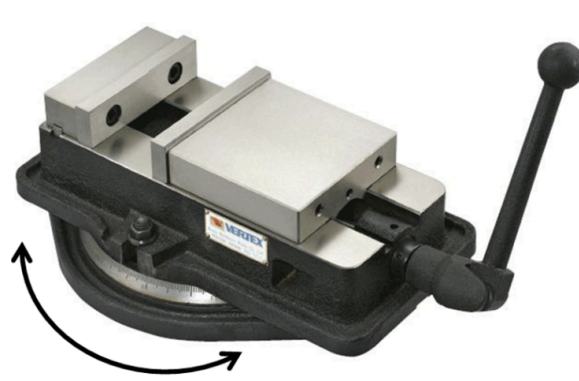
holds workpiece w/ flat surfaces.



### Swivel Vice

holds workpiece with flat surfaces.

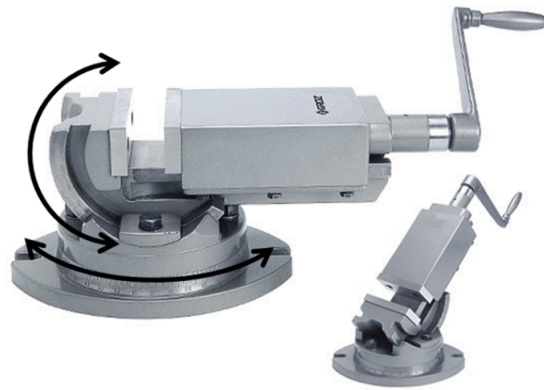
can swivel, adding extra machining axis (vertical).



### Universal Vice

holds workpiece with flat surfaces

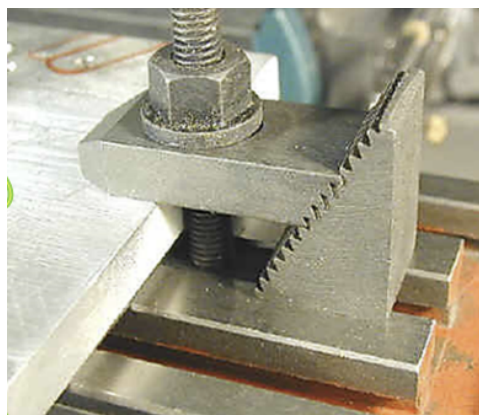
vice can swivel and tilt, adding 2 machining axis, for machining compound angle surfaces.



### Clamps, Tee-Bolts & Nuts (as a whole clamping set)

different clamps are used depending on the work surface, like bulky or irregular shaped work.

Other than the clamp, a clamping set also consists of a set block and tee bolts/tee nuts.



from left to right, the work, the clamp, the tee bolt, and the step block.

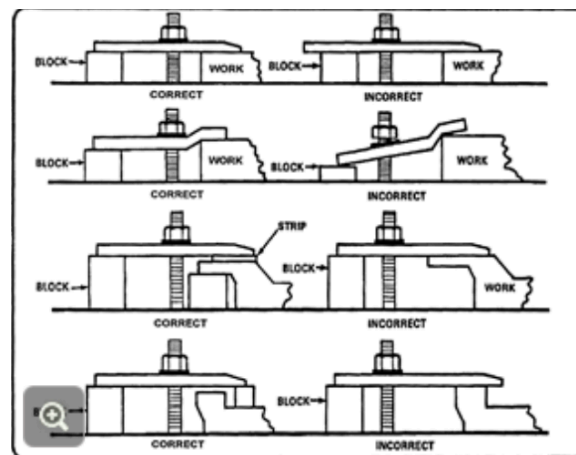
### Proper Clamping

To use clamps properly, the bolt should be close to the work, to maximise clamping force

The clamping force should be downwards only, so that no horizontal component of the clamping force pushes the work away.

Use shims to protect the workpiece, if needed.

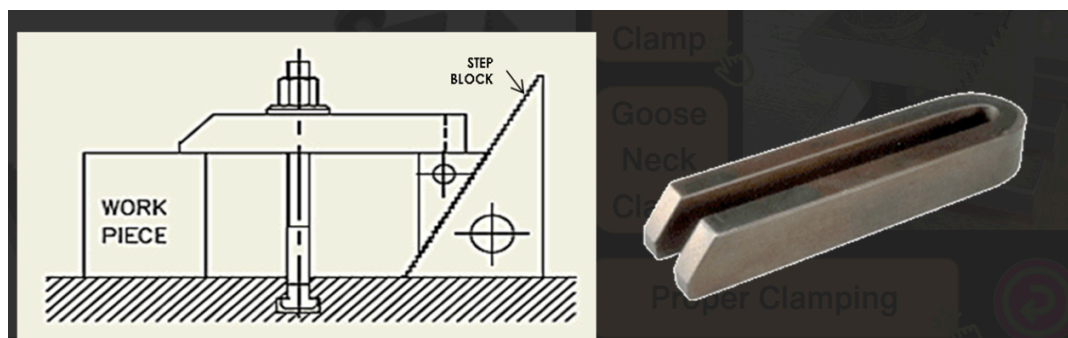
ensure clamping surface is large & as inward to work, to prevent moments and lifting of work.



proper clamping will then allow higher cutting and feed speeds.

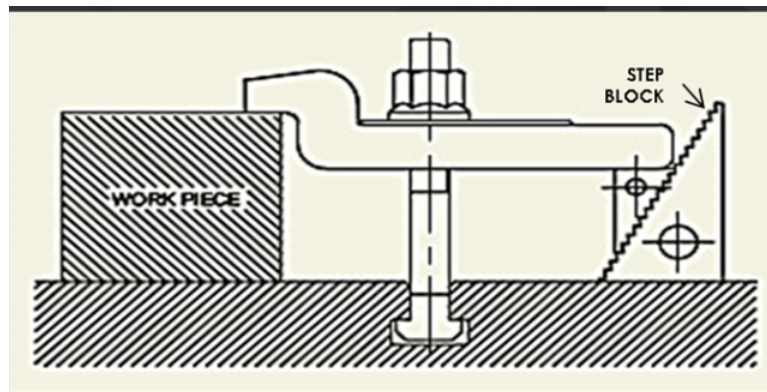
### U Clamp

this clamp can be removed without taking the nut off the tee-bolt, making it useful for quick removal and clamping.



### Goose Neck Clamp

has a lower profile than other clamps, which improves rigidity by lowering clamping distance.



### Jigs & Fixtures

a custom made part to hold a specific workpiece(s), to secure and release them effectively, and to be reused for repetitive work.

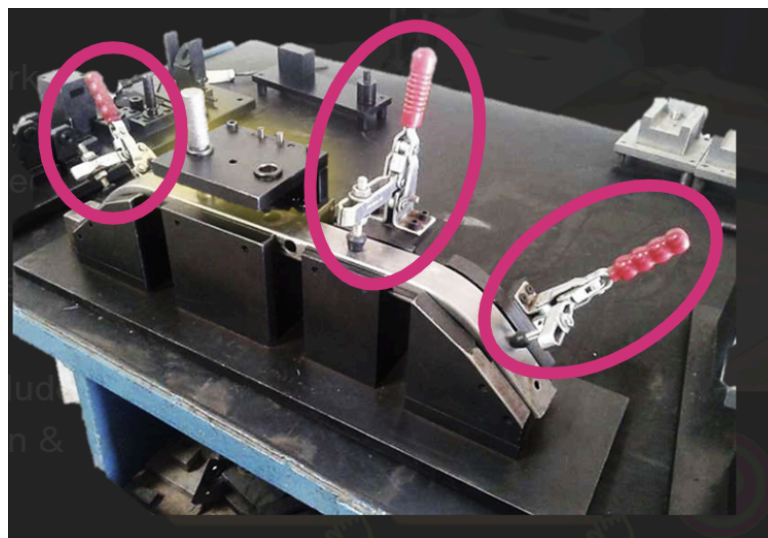
thus these are used for increasing production speed in mass production.

these jigs and fixtures are fixed to the machine via clamps, bolts and nuts.

cutter setting blocks may also be used, for easy setup of cutter location & height

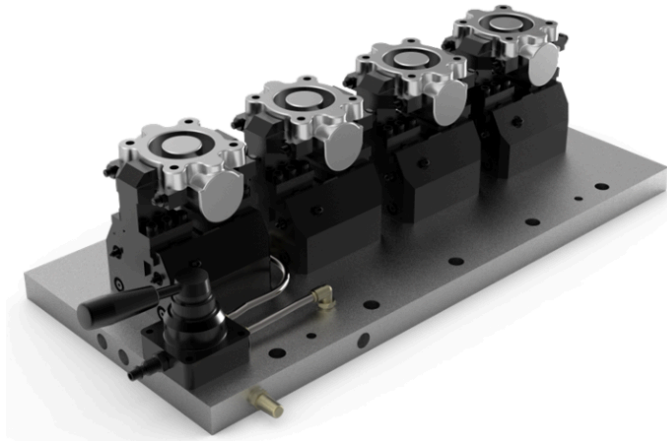
### Jigs

jigs are used to guide a tool to the correct position (as highlighted by the yellow glow), and to locate and hold the work in the correct position.



## Fixtures

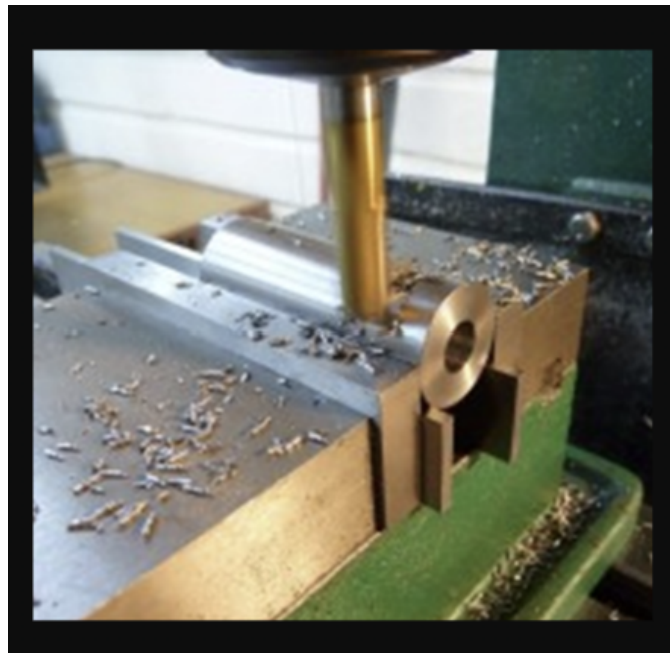
fixtures are used to locate and hold the work in the correct position.



## Parallel Bars

used in combination with a vice.

raises workpiece above machine vice, to prevent tool from cutting into vice, and also clamps workpiece parallel to machine vice.



like clamps, the parallel bars have to be used correctly:

work should be clamped in middle of vice.

work must always be supported by parallel bars, absence of parallel bars may cause work to swivel under cutting forces.

if the work is tall enough, parallel bars are not needed and adding parallel bars will only reduce the clamping area.

